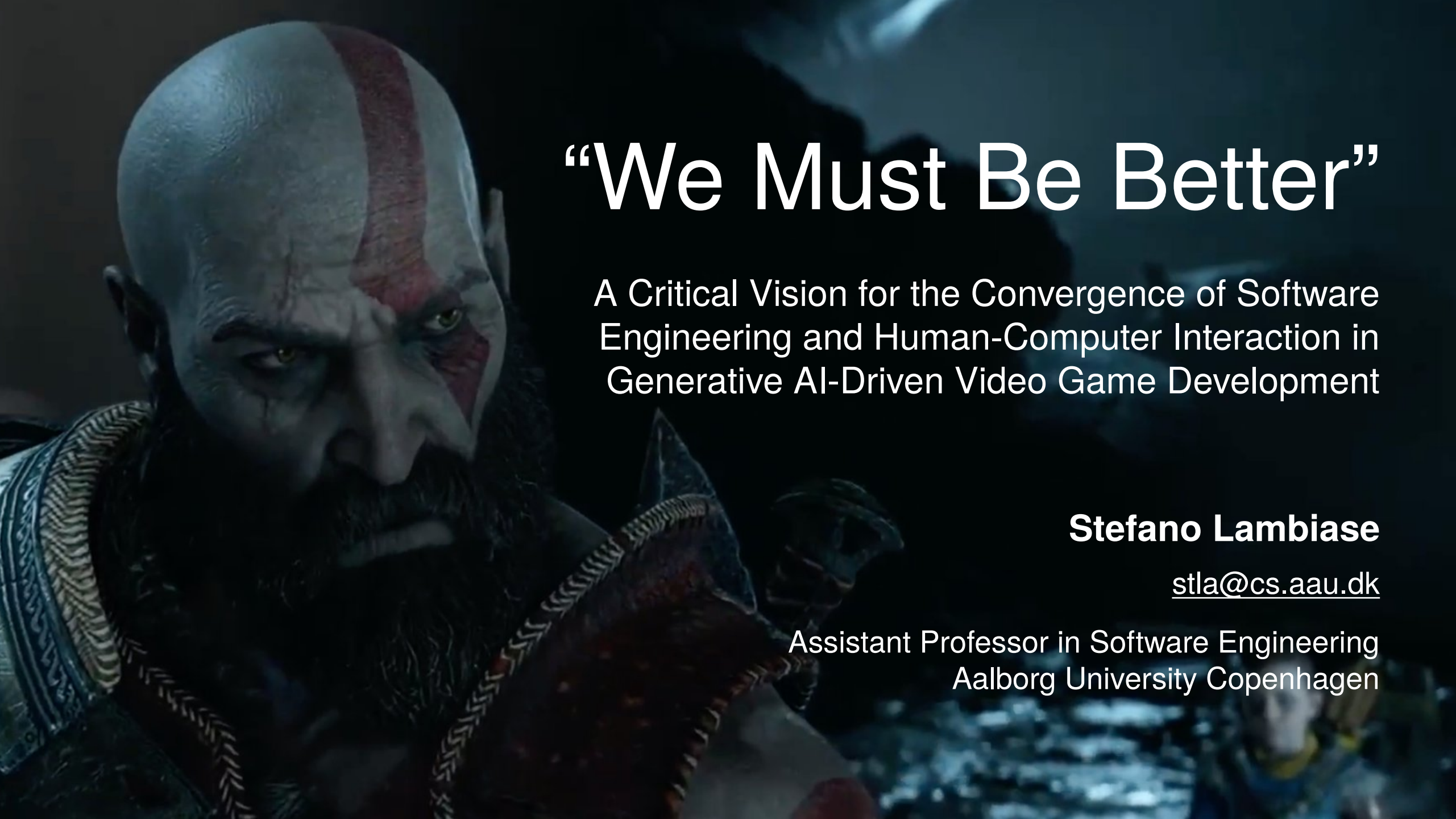




“We Must Be Better”

A Critical Vision for the Convergence of Software Engineering and Human-Computer Interaction in Generative AI-Driven Video Game Development

Stefano Lambiase

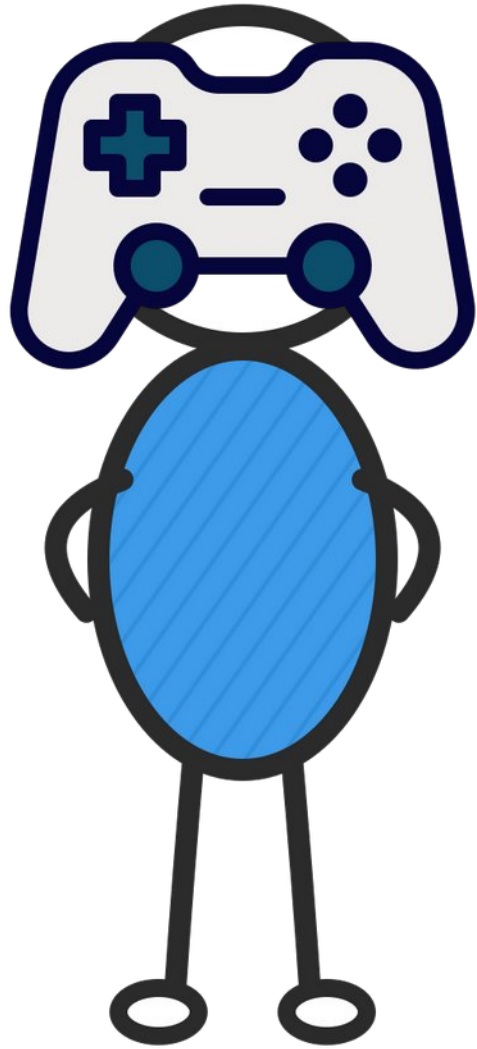
stla@cs.aau.dk

Assistant Professor in Software Engineering
Aalborg University Copenhagen

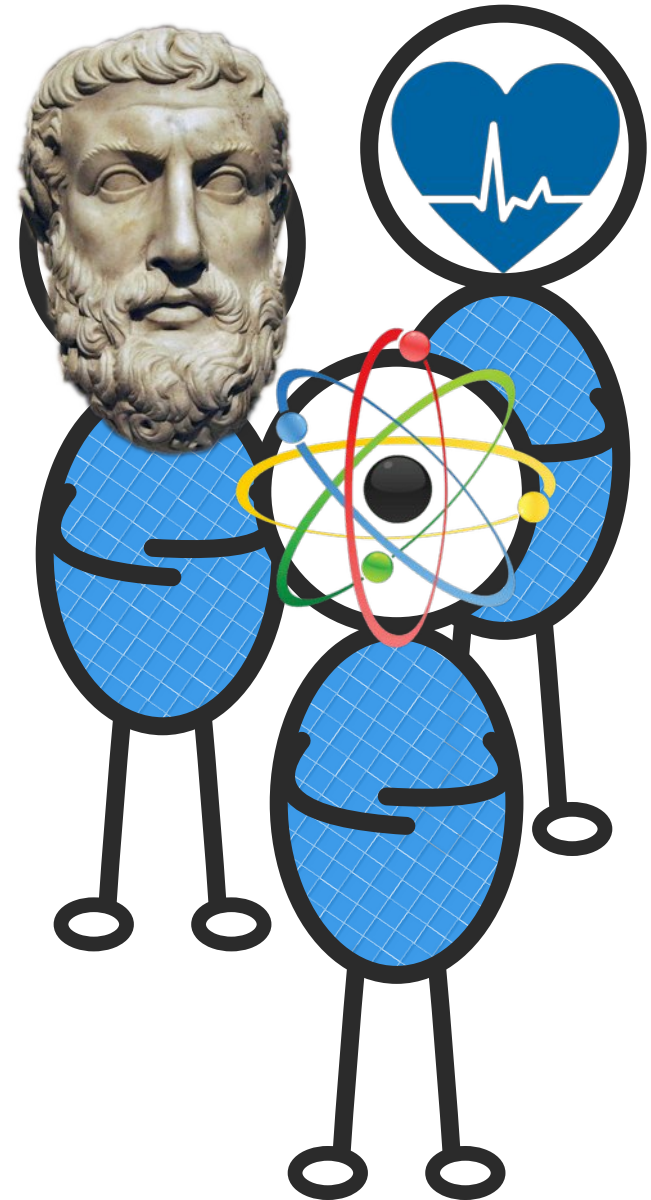


Once upon a time...

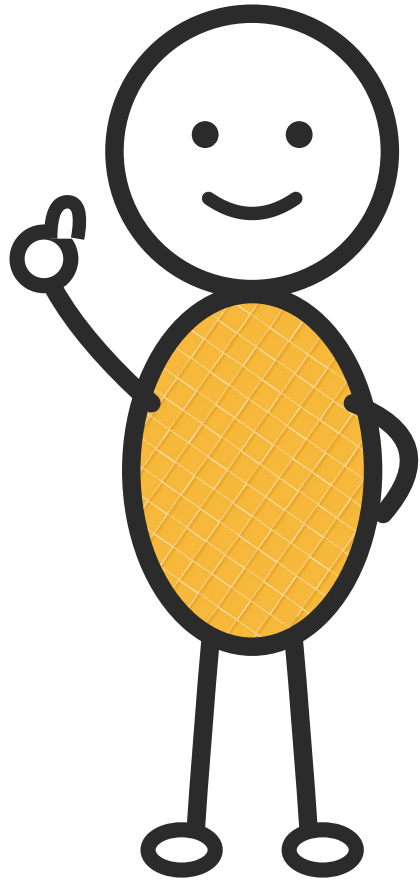




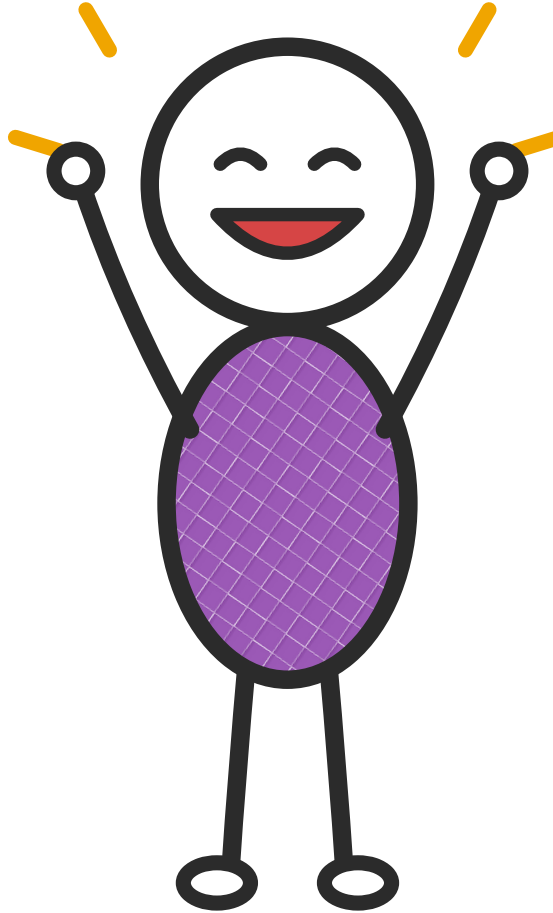
***Game Research**
had a lot of siblings
who were jealous
that it was
researching fun
things, so they
started isolating it.*



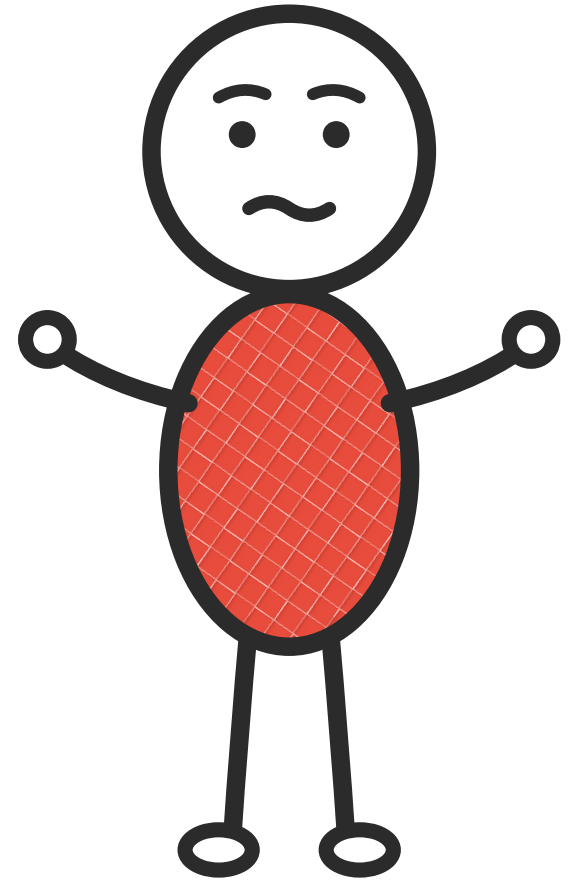
Since it was feeling sad, Game decided to split itself into three so to play to its own...



HCI Game

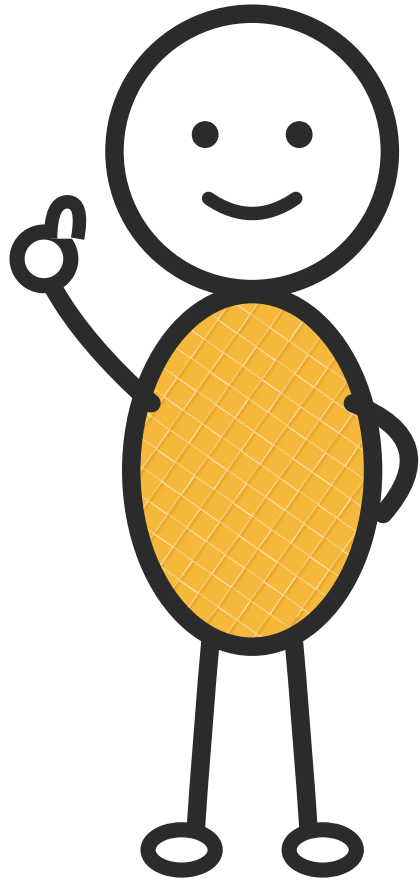


Game Game

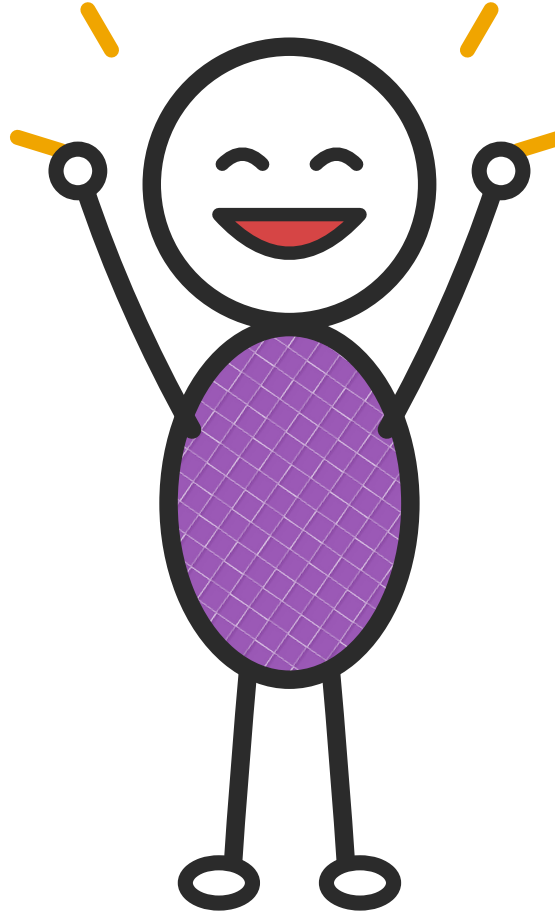


SE Game

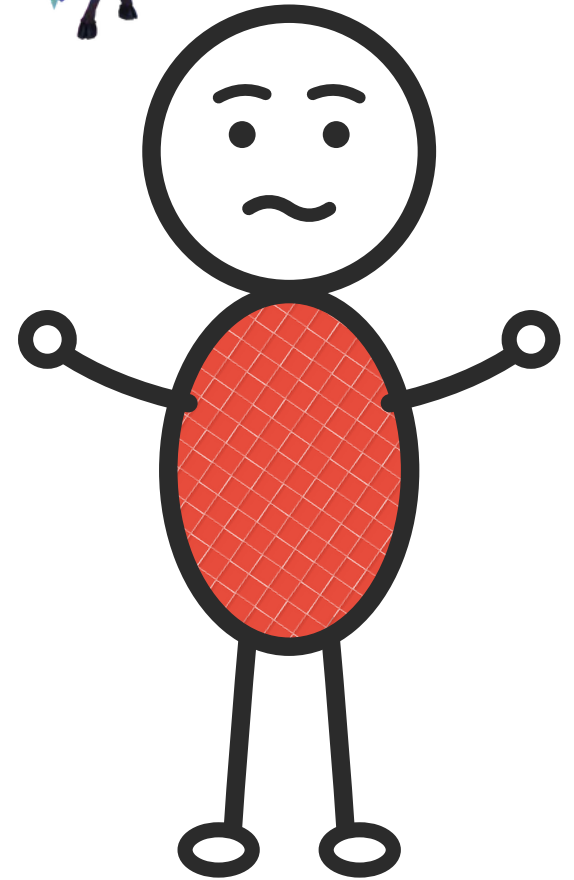
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HCI Game



Game Game

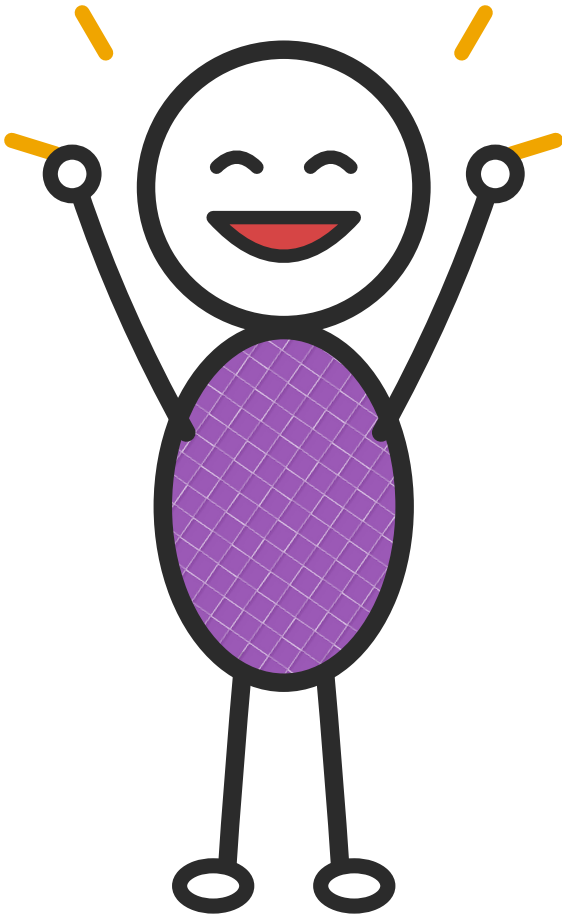


SE Game

A cartoon character with a white head, a smiling face, and a yellow body with a black grid pattern. The character is waving with its right hand.

Figure 1. Display format for the Darts game.

Game Game, like its sister publication, was beginning to establish itself as the main and most diverse field... With publications at various levels starting from the first 2000



Game Game

Game Studies

the international journal of computer game research

volume 1, issue 1
July 2001

home about archive

PLAYER ONE

Espen Aarseth is Associate Professor in Humanistic Informatics at the University of Bergen. His best-known book is *Cybertext: Perspectives on Ergodic Literature* (1997). Aarseth has studied computer games since 1984. He is the founder of the Digital Arts and Culture annual series of international conferences (1998 -), and has directed the Lingo project at the University of Bergen (1997-2000), using and developing MOOs for language learning.

Computer Game Studies, Year One
by Espen Aarseth, Editor-in-Chief

Welcome to the first issue of the dedicated to computer game studies and perhaps the most remarkable has been started before. As we know for almost as long as there have been the first modern game, turns for the genre has existed for three decades before?

2001 can be seen as the **Year** emerging, viable, international first international scholarly conference Copenhagen in March, and several the academic year when regular game studies are offered for the first time scholars and students seriously, as a cultural field which.

To some of us, computer game cultural importance than, say from 2001, the potential cultural future is practically unfathomal especially multi-player games, a way the old mass media, such novels never could. The old mass shared values and sustained in communities remained *imagined* little or no direct communication player games are not like that. *Quake Arena*, the aesthetic and this could be regarded as the game since the invention of the computer games as merely the as some do, is to disregard the force outdated paradigms onto considerable Hollywoodisation that started with the "interactive" but there is also a world wide, movement that has a better in movement before it. Hollywood distribution, and now there is a booth: the Internet. Even Bill Gates Internet, and there is much less succeed. From the closed ecosystem games communities on the Net would be a mistake to assume industrial complex will rule, and studies strategy, this would be

A cognitive, communication revolution?
Much hype has been produced instigate new ways of thought which was supposed to give us to the way our brains worked, communication. So far, however

Game Studies

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Markku Eskelinen is an independent scholar and experimental writer of ergodic prose, interactive drama, critical essays and cybertext fiction. He holds a cand. philol. in comparative literature and is now co-editing a series of *Cybertext Year Books* with Raine Koskimaa. Excerpts from his early fiction were published in *The Review of Contemporary Fiction* (Summer 1996) according to which he's also "easily the most iconoclastic figure on the Finnish literary scene."

He has given paper and other presentations at various international conferences, including the series of Digital Arts and Culture conferences and the ACM conferences on Hypertext and Hypermedia.

The Gaming Situation
by Markku Eskelinen

1. Introduction
The first point of departure for Outside academic theory people between narrative, drama and to drop it and wait until it starts games and especially computer almost without exception colon and film studies. Games are seen remediated cinema (1). On top being successful in terms of information grounded, as they are usually (mainstream drama or outdated seriously and hilariously obsolete commedia dell'arte, Victorian in the scene. To put it less nicely, altogether and avoid any intellectual assassination of both the avant course that in the long run such commercially incorrect.

In any case, in what follows I'll gaming situation by trying to point elementary qualities that set it both of the latter being rather slightly beyond the necessary have to enter in order to gain it. Historically speaking this is a both attractions, practices and very on, not to mention lots of money.

As we study computer games, well as of games. For that purpose Roger Callois, Warren Motte through which the possibly neighbouring disciplines and modified and transformed. While functionality, typology and orient the bare essentials of the game configuration of temporal, spatial in different registers.

2. Gaming as configuration
Regarding the so-called remediated possible statement would be that not presentations or narratives. Parlett formal games are systematic latter part consists of specific points

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Jesper Juul is a Ph.D. student in computer games at the IT University of Copenhagen. He holds an M.A. in Nordic Literature with a master's thesis on interactive fiction. He co-arranged the Computer Games & Digital Textualities conference in March 2001. He also develops chat and multiplayer games in the company Soup.dk.

Games Telling stories? (1)
-A brief note on games and narratives
by Jesper Juul

Introduction
As questions go, this is not a bad one: Do games tell stories? Answering this should tell us both *how* to study games and *who* should study them. The affirmative answer suggests that games are easily studied from within existing paradigms. The negative implies that we must start afresh.

But the answer depends, of course, on how you define any of the words involved. In this article, I will be examining some of the different ways to discuss this. Let this turn into a battle of words (i.e. who has the right to define "narrative"), my agenda is not to save or protect any specific term, the basic point of this article is rather that we should allow ourselves to make distinctions.

The operation of framing something as something else works by taking some notions of the source domain (narratives) and applying them to the target domain (games). This is not neutral; it emphasises some traits and suppresses others. Unlike this, the act of *comparing* furthers the understanding of differences and similarities, and may bare hidden assumptions.

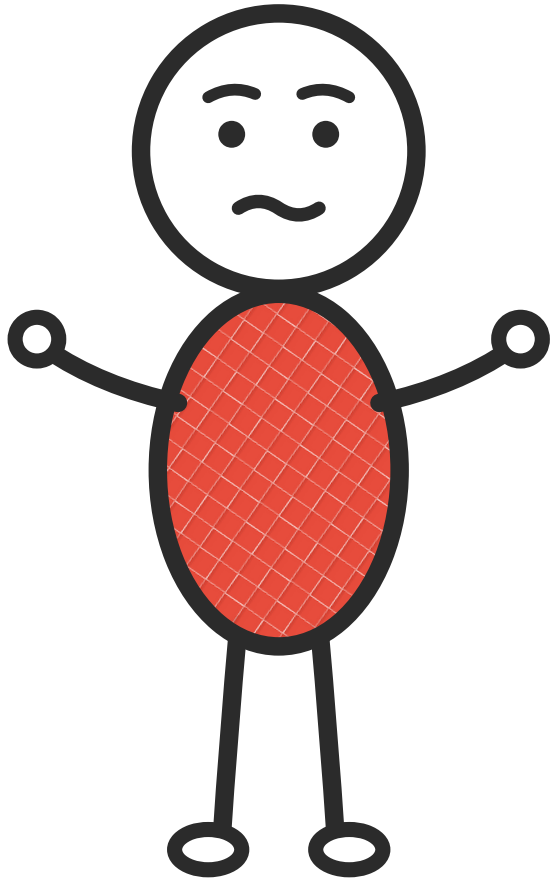
The article begins by examining some standard arguments for games being narrative. There are at least three common arguments: 1) We use narratives for everything. 2) Most games feature narrative introductions and back-stories. 3) Games share some traits with narratives.

The article then explores three important reasons for describing games as being non-narrative: 1) Games are not part of the narrative media ecology formed by movies, novels, and theatre. 2) Time in games works differently than in narratives. 3) The relation between the reader/viewer and the story world is different than the relation between the player and the game world.

The article works with fairly traditional definitions of stories and narratives, so as a final point I will consider whether various experimental narratives of the 20th century can in some reconcile games and narratives.

Telling stories
Everything is narrative / Everything can be presented

The third brother is the one who's a little slower... Most of the important publications started in 2009...



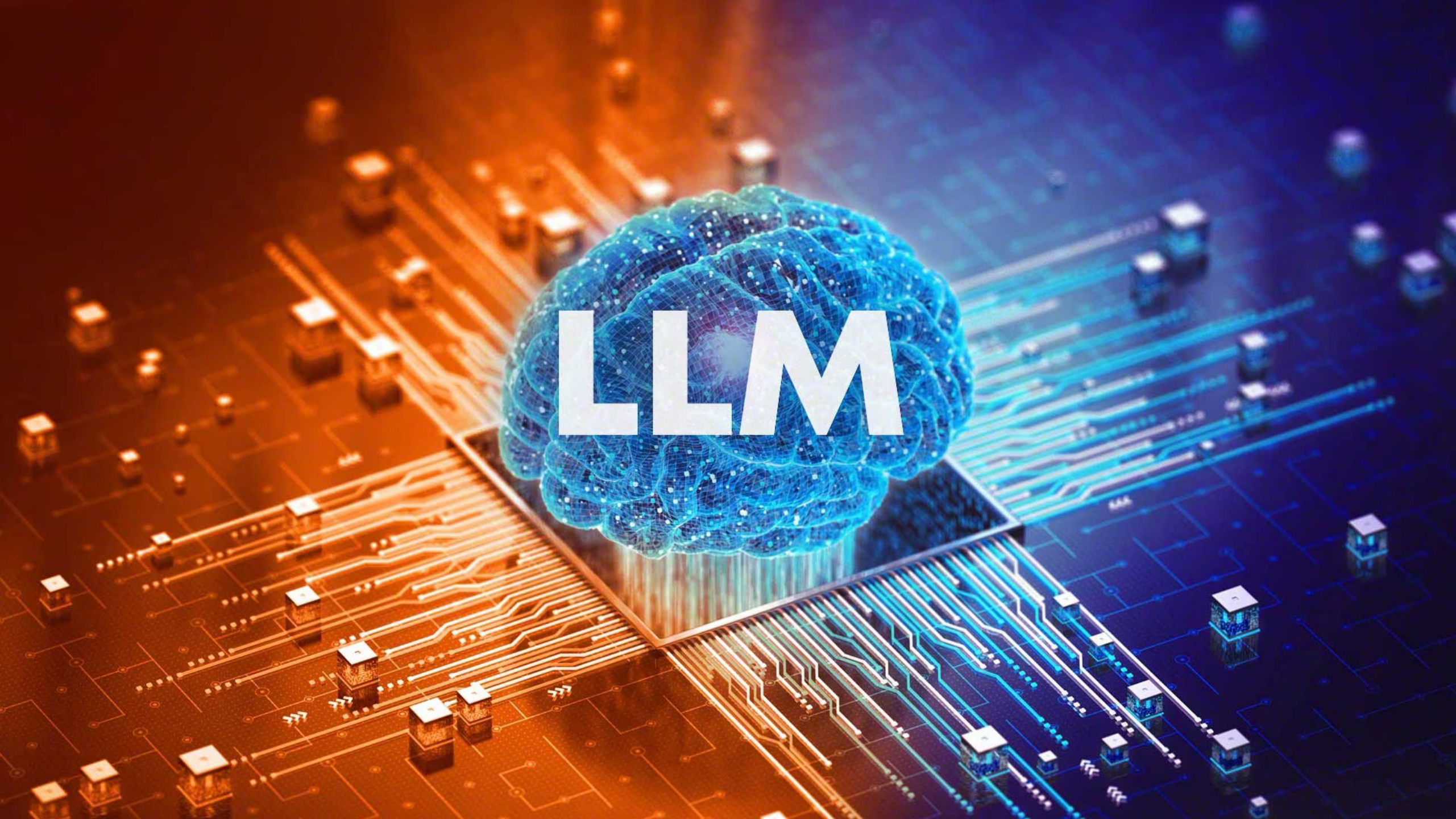
SE Game





How much do you agree with the story
above?



The image features a glowing blue, wireframe brain with a mesh-like texture, positioned centrally. Overlaid on the brain are the large, white, bold letters "LLM". The background is a detailed circuit board with intricate gold-colored traces and various electronic components. The left side of the board is illuminated with a warm orange glow, while the right side transitions into a cool blue light. The overall composition suggests a connection between artificial intelligence and computer hardware.

LLM

What if we
integrate GenAI
Agents into video
games to animate
non-player
characters?





**It will be possible to
make these rough and
clumsy NPCs...**

Dynamic





Resilient to the unexpected



Adaptable

Realistic



A person is shown in profile, wearing a VR headset and large headphones, focused on a game displayed on a large, curved monitor. The game features a vibrant, colorful environment with trees and structures. The person's hands are visible, typing on a backlit keyboard. The scene is dimly lit, with the primary light source being the monitor and the ambient glow of the game.

Improve gamers
experience

A photograph of a woman and two children playing in a swimming pool. The woman is on the right, smiling and holding the hands of a young girl in the center. The girl is also smiling. In the foreground, the head of another child is visible, looking up. The pool is blue, and the background shows a chain-link fence, palm trees, and a building under a clear sky.

HCI

GenAI evaluation
methodology

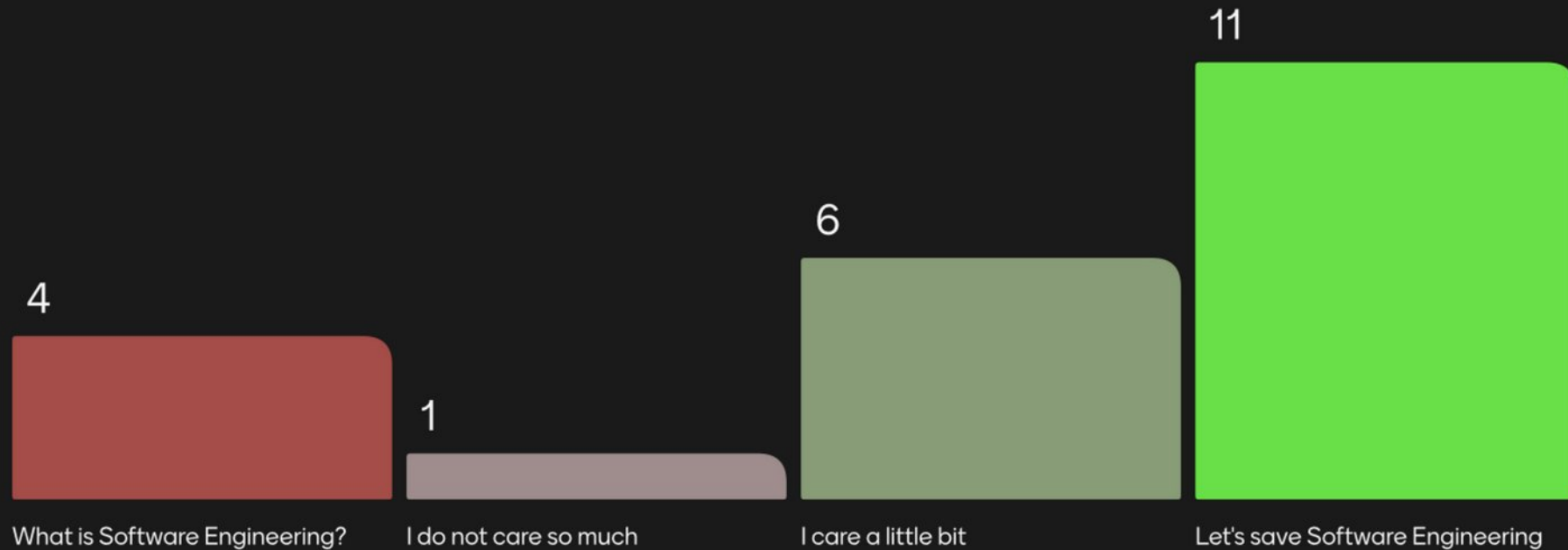
Game industry
& games venues

A full-page background image showing a human skeleton sitting on a wooden chair underwater. The water is a murky teal color, and the skeleton is positioned centrally. The text 'Software Engineering' is overlaid in white, bold font across the middle of the image.

Software Engineering

Who deserves the lifeguard first? Rank from "save them now" (top) to "they'll be fine" (bottom)

Do we care about the poor destiny of Software Engineering research in games?



menti.com
4912 5026

22 of 30 responded



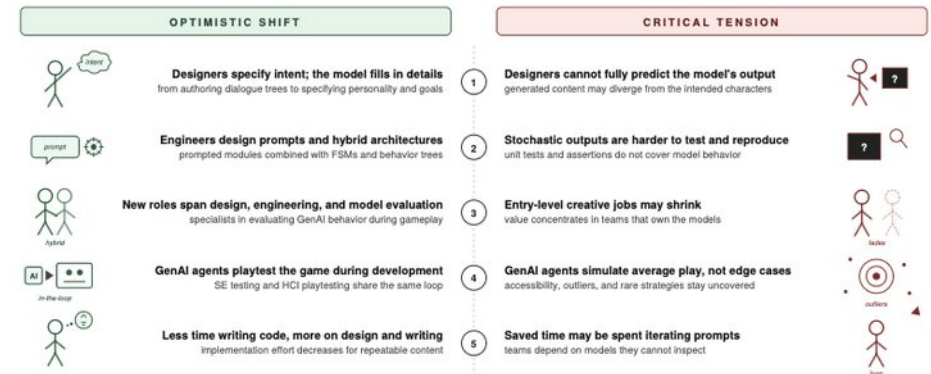
“We Must Be Better”

A critical review of the literature to identify current weaknesses in SE research in Games and proposes a vision for the non-disruptive integration of generative AI into video game development teams.

“We Must Be Better” A Critical Vision for the Convergence of Software Engineering and Human-Computer Interaction in Generative AI-Driven Video Game Development

Stefano Lambiase
Aalborg University Copenhagen
Copenhagen, Denmark
stla@cs.aau.dk

Five shifts in GenAI-driven game development



Abstract—Generative AI has the potential to profoundly change video game development, but its impact extends beyond the product to the process itself, opening the possibility of transforming the sector and making it even more creative than it already is. Three research traditions—i.e., game software engineering, generative AI in games, and simulated users in software testing—converge on the same object of study but operate in disciplinary silos with limited cross-fertilization. I argue that this is a missed opportunity, given the risk of failing to develop a shared vocabulary and of producing partial, scattered,

and non-transferable knowledge. I envision, optimistically, that if leveraged carefully, generative AI can be the moment that brings game research to convergence, mirroring in the research community the transdisciplinary integration that already characterizes industrial practice. To discuss this, the paper adopts the critical review methodology, extended to generate a constructive vision: a shift that redefines designer and engineer roles around prompt and constraint design, opens new hybrid professional roles, and converges software engineering testing with Human-Computer Interaction playtesting. I argue this has the potential to reshape the field and to open extensive research lines, both on the constructive side and on the structural risks the shift raises.

Index Terms—video game, software engineering, generative AI, large language models, testing, human-computer interaction

I. INTRODUCTION

In *God of War*, Kratos turns to his son Atreus and says “We must be better” [1]. The line captures a stance that combines honest acknowledgment of past failure with the obligation to act differently going forward, without giving in to cynicism

“We Must Be Better”

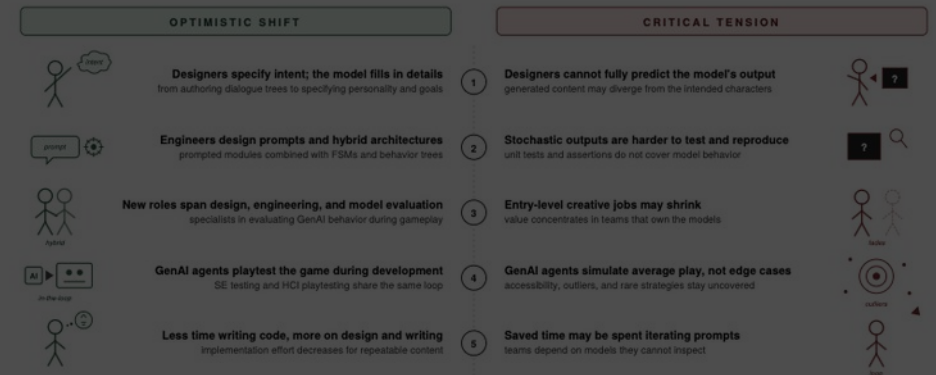
A critical review of the literature to identify current weaknesses in SE research in games and proposes a vision for the non-disruptive integration of generative AI into video game development teams.

WHY?

“We Must Be Better” A Critical Vision for the Convergence of Software Engineering and Human-Computer Interaction in Generative AI-Driven Video Game Development

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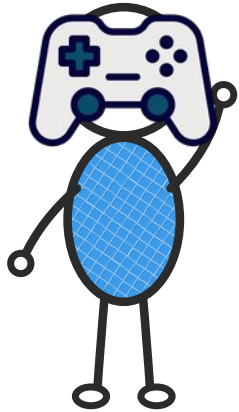


Another story...



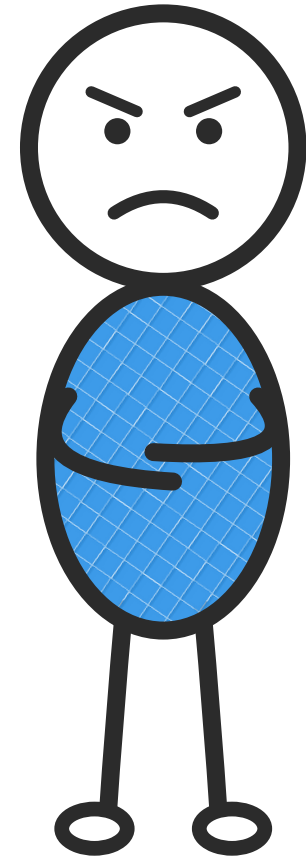
Prologue

The Authored Game

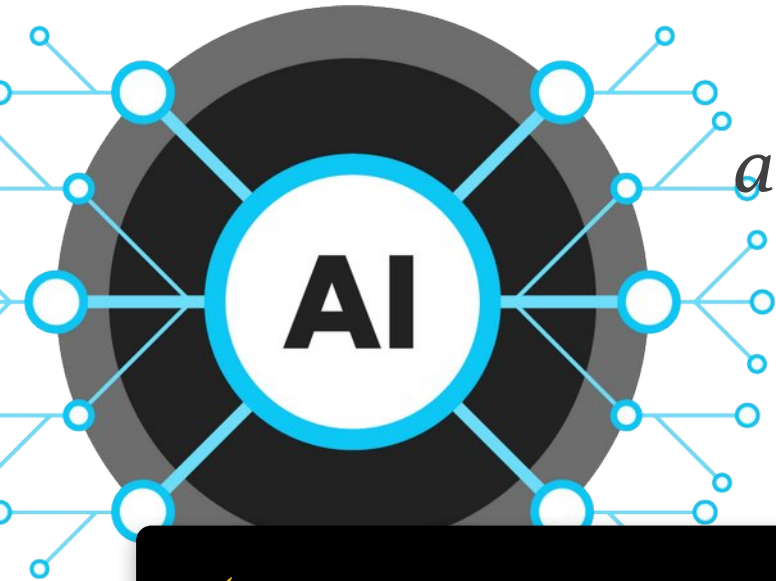


*“I just do what my parents
taught me to do. I don’t make
anything up! I’m dull and cold!”*

— A typical NPC

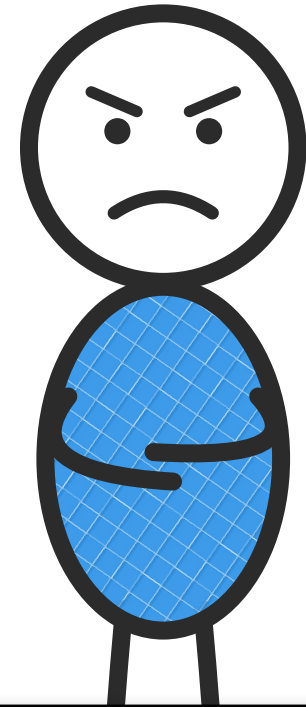


The Authored Game



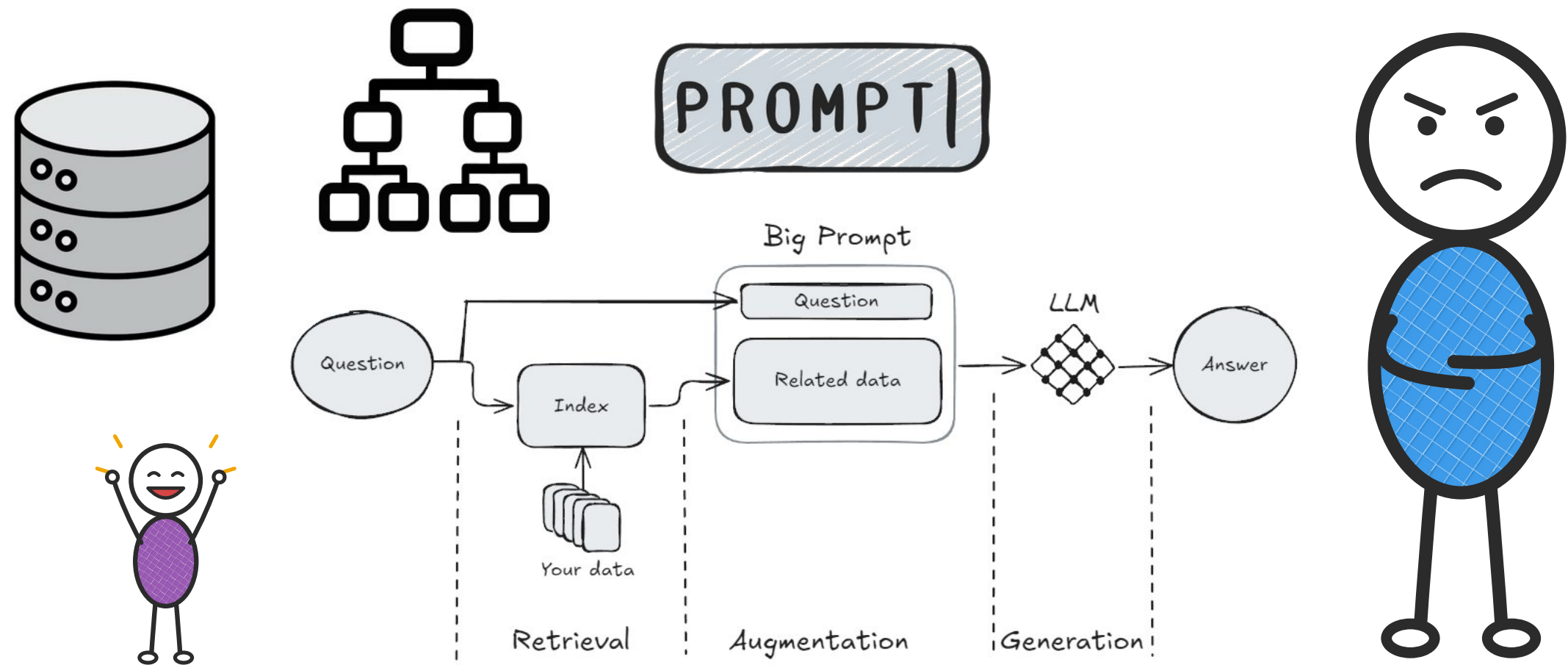
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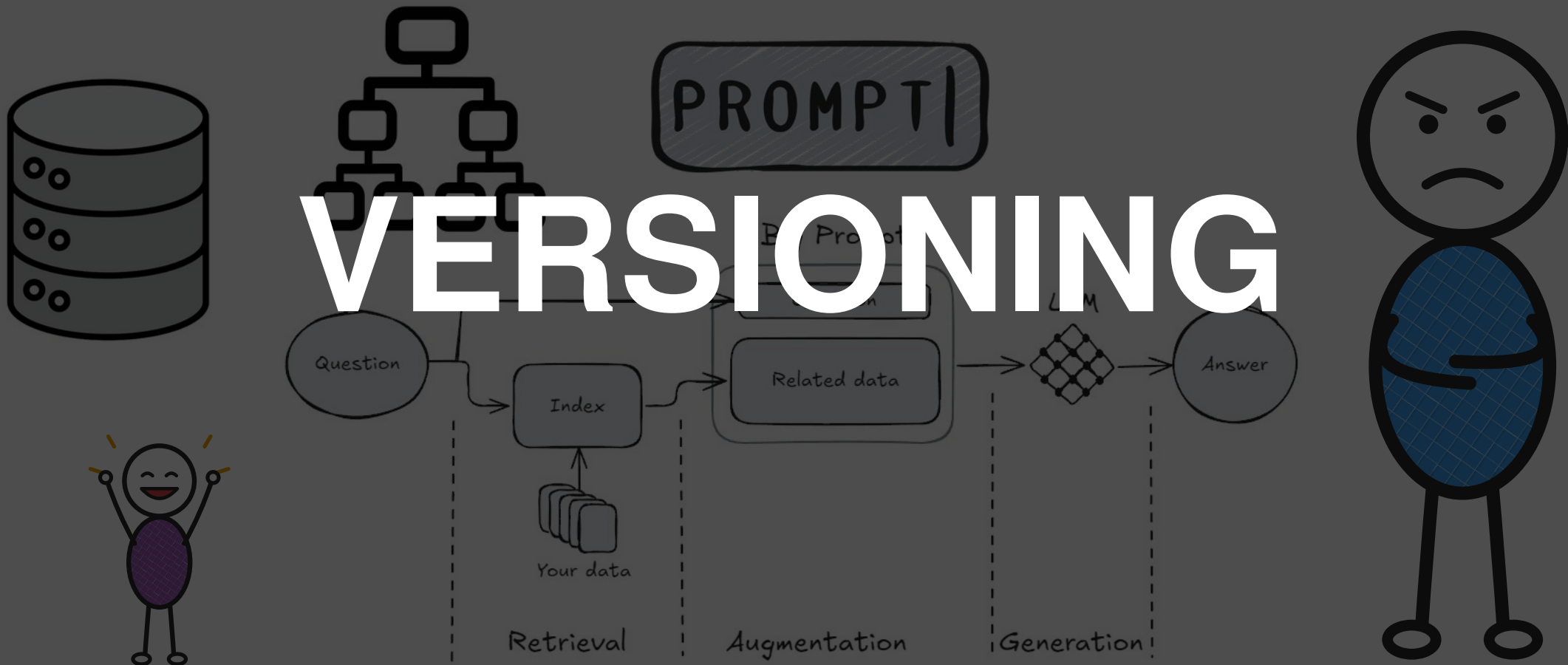


⚡ **THE TRIGGER**—GenAI agents arrive. Unpredictable, emergent, co-created artifacts enter the pipeline—and the story begins.

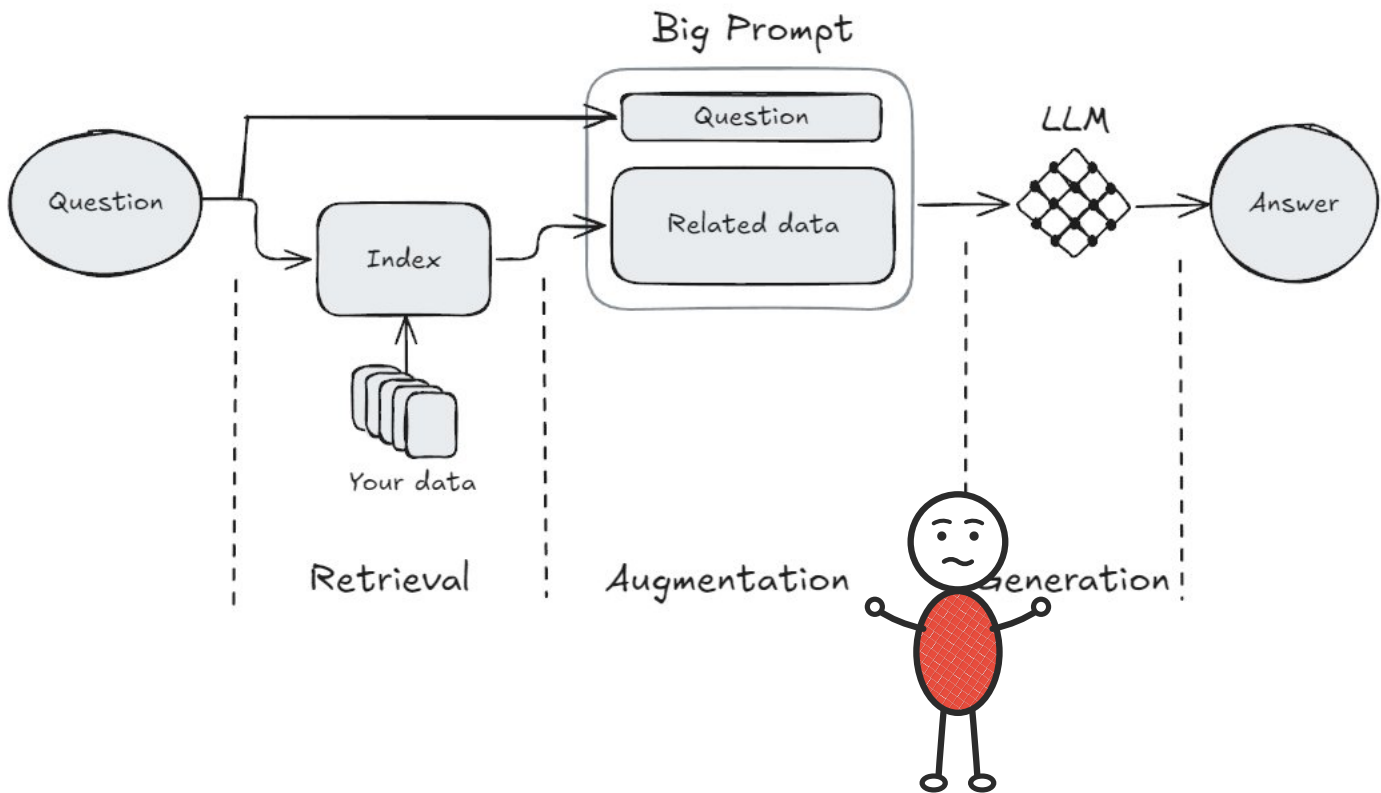
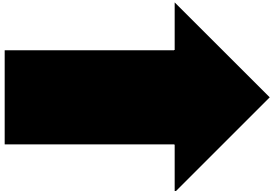
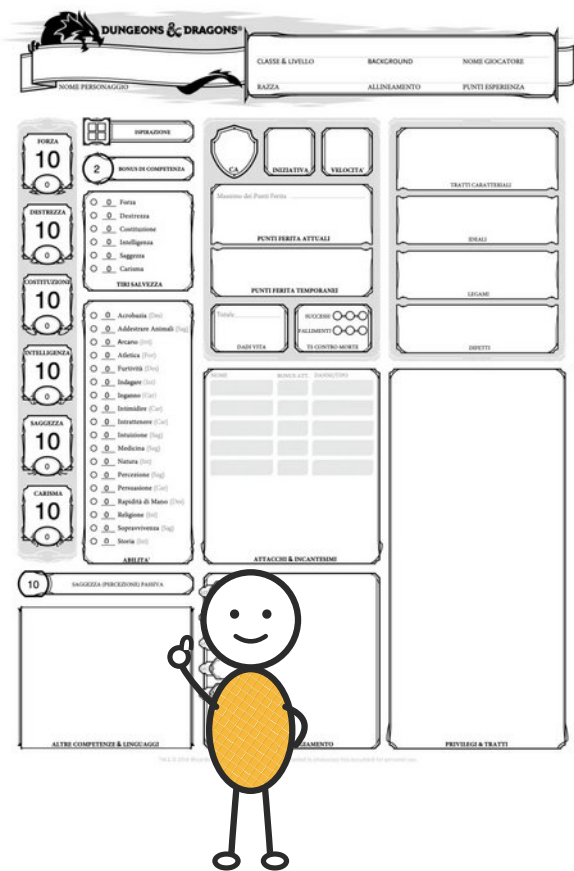
3.A—GenAI as Substrate for New Game Artifacts

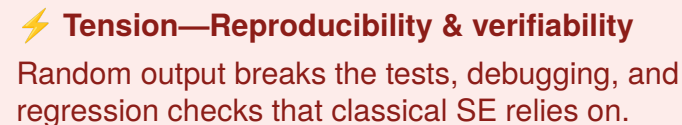
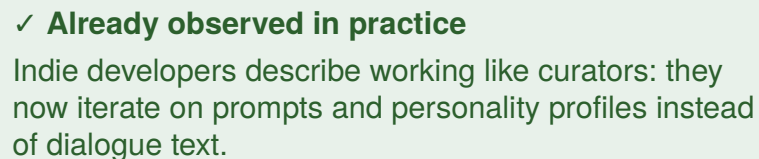


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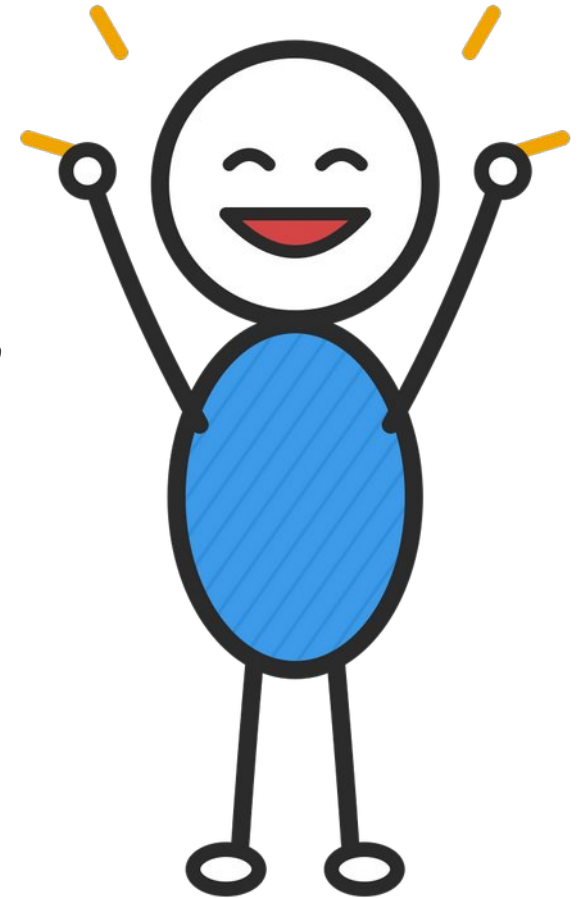
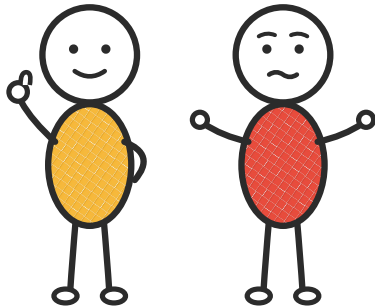




3.B–C—The Shift in the Designer’s and Engineer’s Role

*“Wow! I have a personality, but
I was taught to keep my
composure!”*

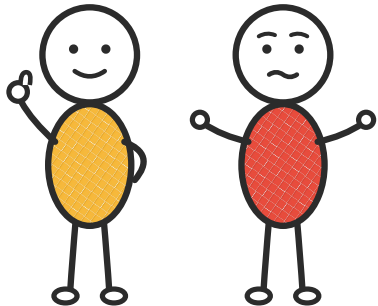
— a GenAI-enabled NPC



3.B–C—The Shift in the Designer’s and Engineer’s Role

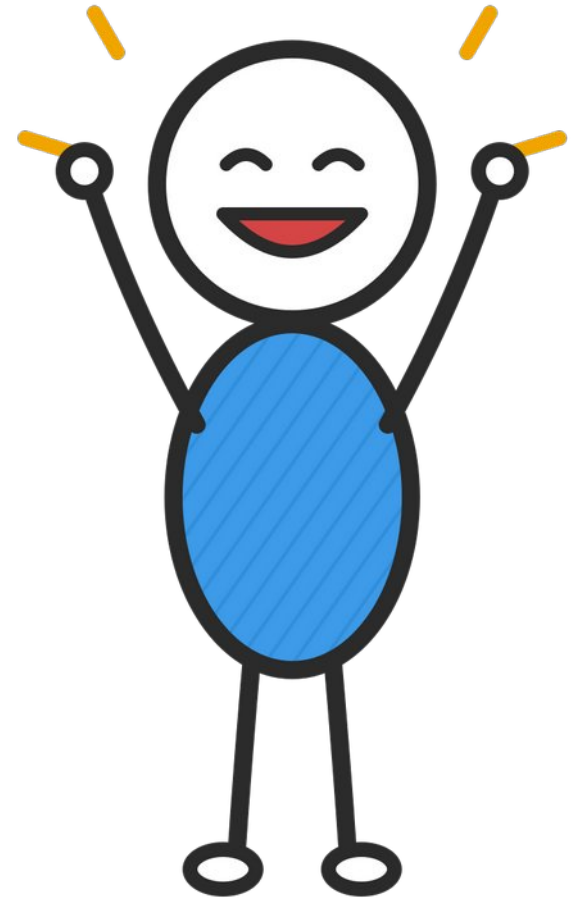
⚡ **Tension—Accountability & opacity**

Because content is co-created at runtime, decisions are partly opaque: when an NPC says something offensive or wrong, it is hard to say who is responsible.



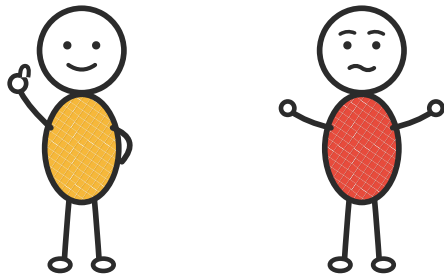
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3.D—Generative Playtesting: the SE–HCI Convergence

GenAI agents become a shared base for SE-style coverage testing and HCI-style playtesting—inside the same development loop.



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GenAI agents become a shared base for SE-style coverage testing and HCI-style playtesting—inside the same development loop.

Playtest

Test Cases

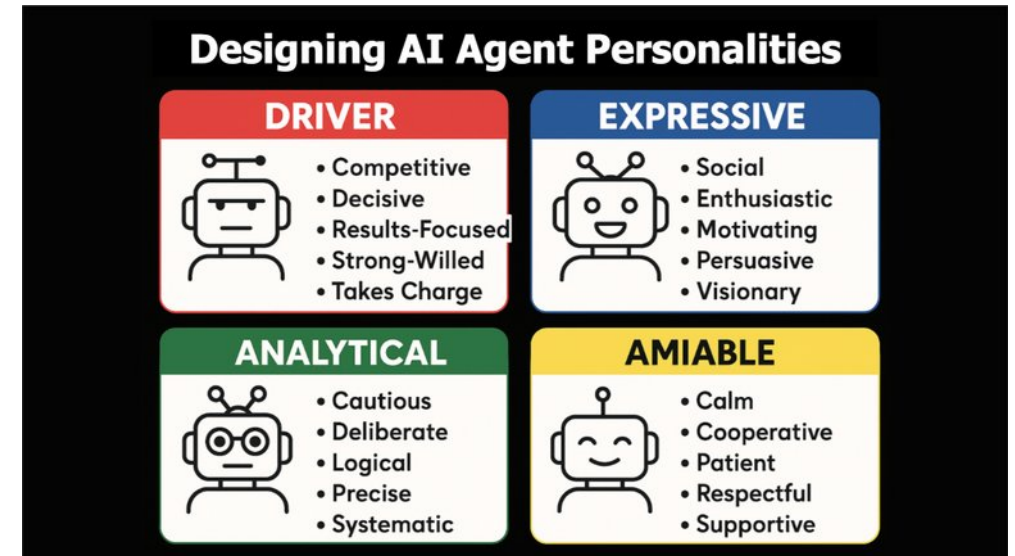


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Playtest

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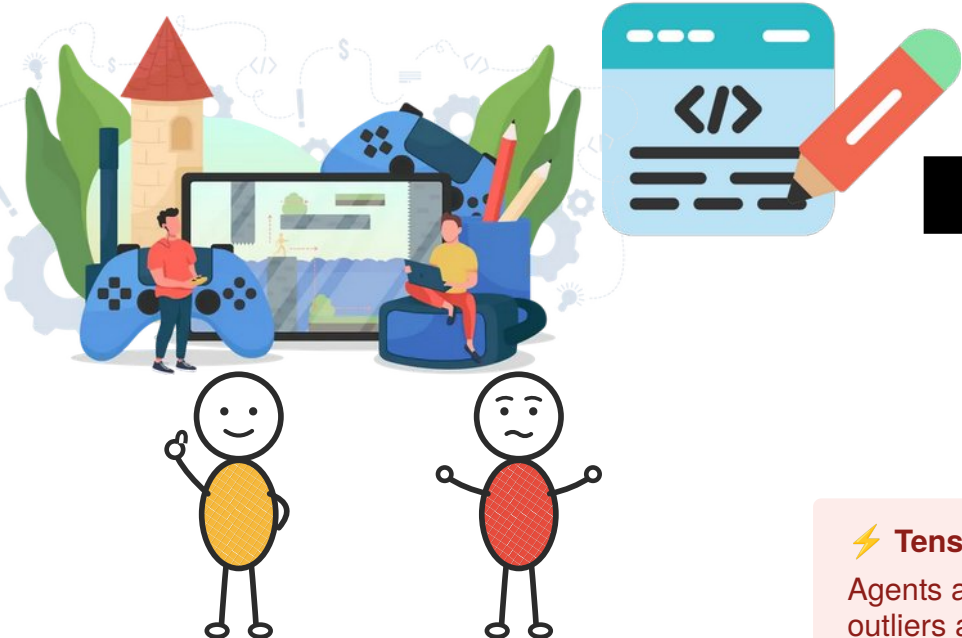
<https://www.linkedin.com/pulse/designing-ai-agent-personalities-matthew-a-mattson-esq--9ibpe/>

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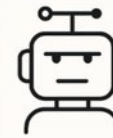
Playtest

Test Cases



Designing AI Agent Personalities

DRIVER



- Competitive
- Decisive
- Results-Focused
- Strong-Willed
- Takes Charge

EXPRESSIVE



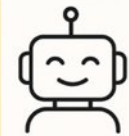
- Social
- Enthusiastic
- Motivating
- Persuasive
- Visionary

ANALYTICAL



- Cautious
- Deliberate
- Logical
- Precise
- Systematic

AMIALE



- Calm
- Cooperative
- Patient
- Respectful
- Supportive

⚡ Tension—False confidence

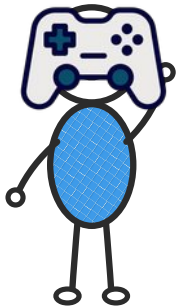
Agents are right on average but miss edge cases—outliers and accessibility.

⚡ Tension—Cost & sustainability

Running models at scale is expensive, and not everyone can afford it; big studios benefit most.

3.E—Hybrid Roles and the Redistribution of Labor

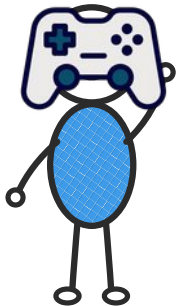
These shifts don't just rename roles—they reshape the work and give hybrid professionals a home.



3.E—Hybrid Roles and the Redistribution of Labor

These shifts don't just rename roles—they reshape the work and give hybrid professionals a home.

- Narrative-AI specialists;
- agent-evaluation specialists;
- constraint architects;
- designer–engineer hybrids
- ...



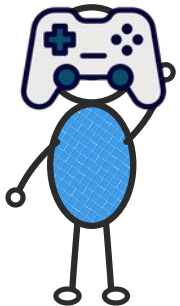
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- ...

⚡ Tension—Deskilling & value redistribution

Companies can still fire anyone because none likes to deal with humans



The Convergent Generative Game

One loop

Strong collaboration between SE and HCI to support developing games for other purposes (e.g., health).



One community

Same language and same knowledge base to support the research community and collaboration.

The background image is a close-up of a dark, spiky, metallic object, possibly a piece of armor or a weapon. It has a complex, layered structure with many sharp, pointed edges. In the background, there is a glowing orange light source, possibly a fire or a lava flow, which creates a warm, fiery atmosphere. The overall color palette is dark with a strong orange glow from the light source.

Epilogue

The Convergent Generative Game

A dramatic landscape featuring a giant stone figure holding a massive sphere, with a city at its base and a sunset sky.

Amazing video games

AI
Agents

Software Engineers

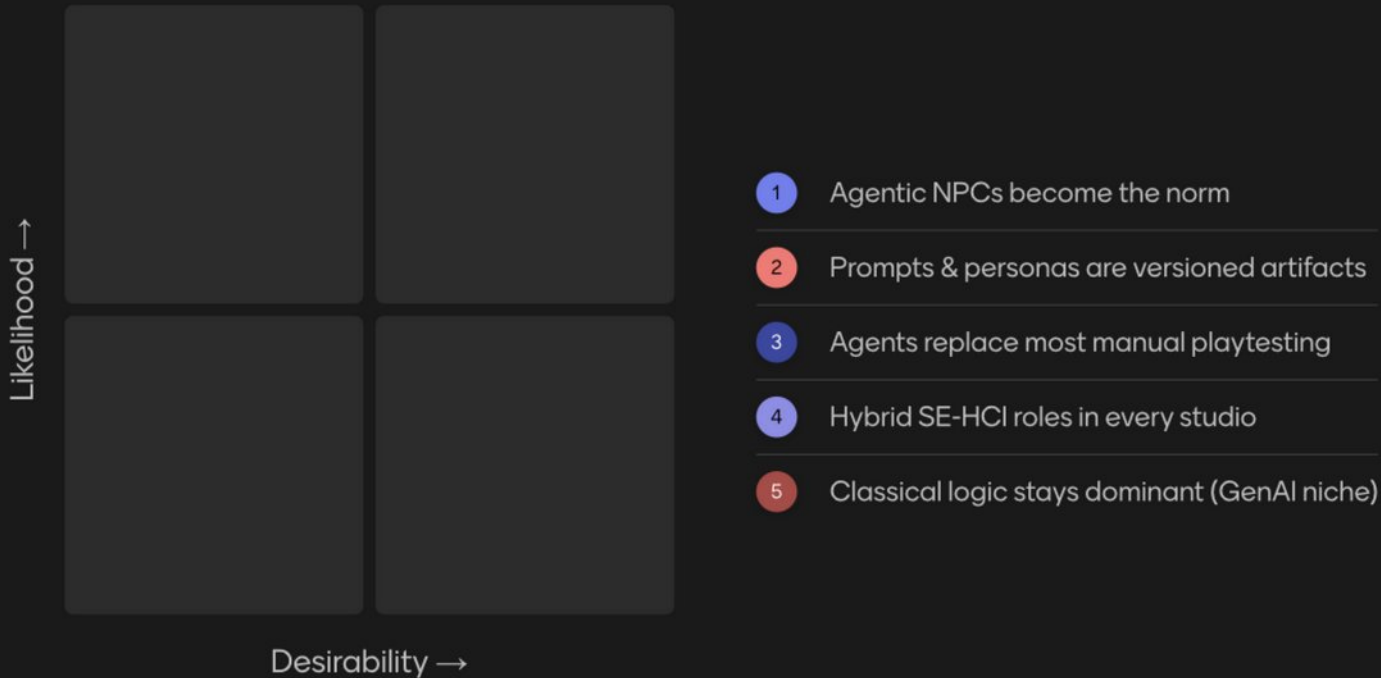
A dramatic scene of a warrior with long red hair, seen from behind, fighting a massive, towering fire. The warrior is holding a sword and has their arms raised in a gesture of defiance or combat. The fire is intense, with bright yellow and orange flames reaching high into the air, creating a sense of scale and danger. The background is filled with smoke and falling embers, suggesting a battle or a catastrophic event.

**AI
Agents**

**Software
Engineers**

EVERYTHING

Which of these futures do you expect—and which do you want?



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0 of 31 responded

Responses are hidden X

Aren't we
forgetting
someone?



Ok, amazing...
but what about the
researchers?



AALBORG
UNIVERSITY



Ok, amazing...
but what about the
researchers?



AALBORG
UNIVERSITY





Research Questions!



**Research
Questions!**



RQ1—Under what conditions do generative AI player simulators reproduce the variance of human player behavior on subjective game qualities within domain-specific tolerance bounds?



RQ2—How can we implement agent testing in a way that fit the needs and budget of real studio workflows?

**RQ3—Do agents with
personalities behave more like
real players, beyond covering
more of the game?**



RQ4—What do AI playtesters miss, and do smaller/cheaper models miss more?





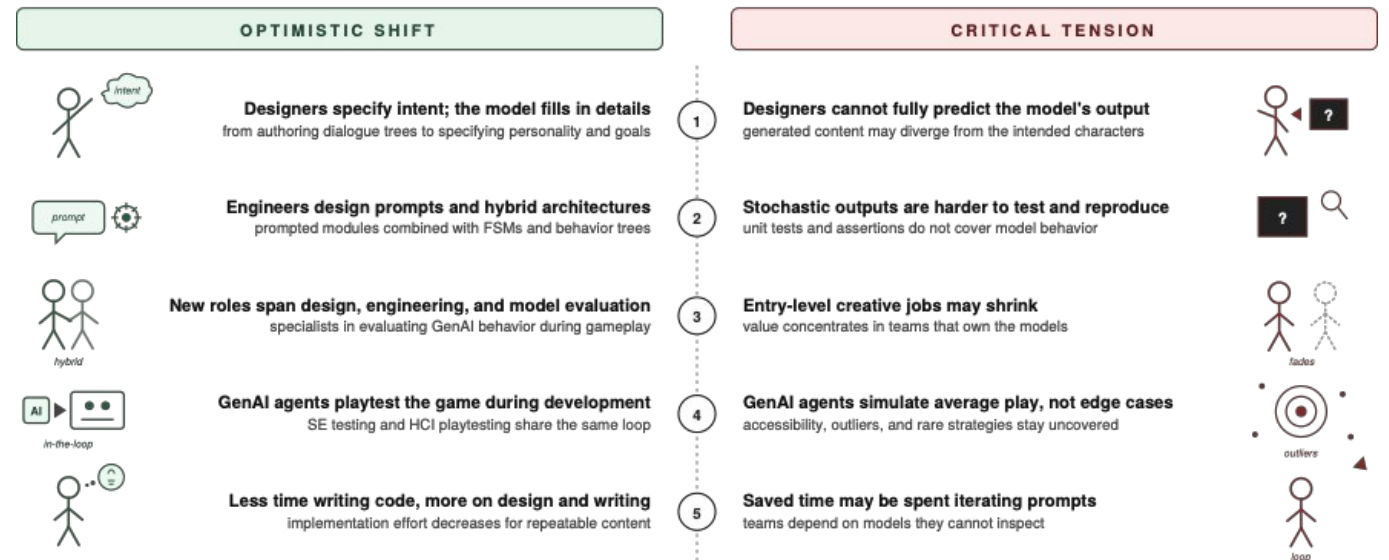
**RQ5—How do studios really
reorganize around GenAI agents
—and what new artifacts become
deliverables?**

Get in touch
and read the
preprint



“We Must Be Better”

Five shifts in GenAI-driven game development



Stefano Lambiase
Assistant Professor in Software Engineering
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Get in touch
and read the
preprint

Questions (I will probably reply to them on LinkedIn)



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Backup Slides



From Tensions to Questions

Each tension the vision raised becomes a question research can answer—five moves (Sec. IV-B).

RQ1

Behavioral Fidelity—When do AI players match the spread of human behavior, not just the average?

answers: False confidence in GenAI-driven testing (Sec. III-D)

RQ2

Pipeline Integration—Does agent testing fit real dev workflows—and studio budgets?

answers: Cost & sustainability (Sec. III-D)

RQ3

Personality Conditioning—Do agents with personalities behave more like real players?

answers: Reproducibility & construct validity (Sec. III-C)

RQ4

Limits, Bias, and Cost Trade-offs—What do agents miss—and does going small make it worse?

answers: False confidence · Cost & sustainability (Sec. III-D)

RQ5

Practices and Roles—How do studios reorganize—and what becomes a deliverable?

answers: Deskilling & value redistribution (Sec. III-E)

RQ1—Behavioral Fidelity

RQ1. When do AI player simulators match the spread of human behavior, not just its average?

Why it matters

- Agents already match the average player's difficulty perception (Xiao & Yang)
- The open question: do they match the full range of players?

How to answer

- Studies with human ground truth
- Many different agents, not one model
- Compare distributions, not averages

⚡ **Closes the loop on:** False confidence in GenAI-driven testing (Sec. III-D)

RQ2—Pipeline Integration

RQ2. Can agent testing fit the pace and budget of real studio workflows?

Why it matters

- The anchor works prove the technique—not that it fits day-to-day development
- It is an engineering question and a business one

How to answer

- Case studies on prototype tooling
- Action research with studio partners
- Follow adoption over time

RQ3—Personality Conditioning

RQ3. Do agents with personalities behave more like real players—beyond covering more of the game?

Why it matters

- MIMIC shows personality helps coverage—no one has shown it helps realism
- First we need shared definitions of personality and behavior

How to answer

- Extend MIMIC with human ground truth
- Build common personas and tooling

RQ4—Limits, Bias, and Cost Trade-offs

RQ4. What do AI playtesters miss—and do smaller, cheaper models miss more?

Why it matters

- Known blind spots: outliers and underrepresented players—but no map of where they hurt most
- Small models trade scale for efficiency, and may widen the gap

How to answer

- Map what agents miss, per model size
- Design and test human+agent protocols

RQ5—Practices and Roles

RQ5. How do studios really reorganize around GenAI agents—and what new artifacts become deliverables?

Why it matters

- The new roles and the labor shift are, for now, hypotheses
- Candidate deliverables: prompts, personas, agent populations, evaluation suites

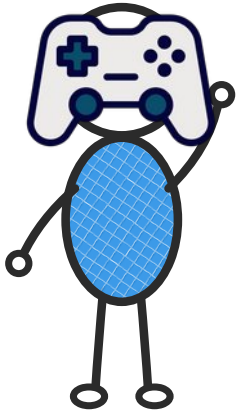
How to answer

- Interviews across studio sizes
- Post-mortems of shipped productions
- Watch teams in transition

Prologue

The Authored Game

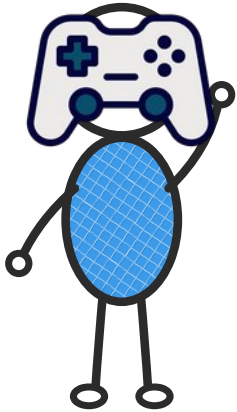
Where the story starts: everything an NPC does comes from something a human designer wrote and an engineer coded.



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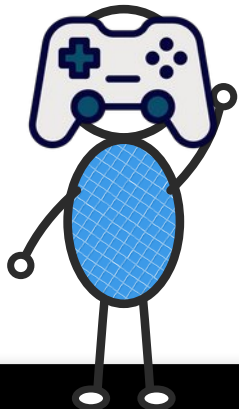


- Designers write the story by hand: dialogue trees, scripted events, fixed quests
- Behavior trees, state machines, and scripts define all behavior before the game ships
- Testing is manual playtesting plus checks with known answers—every run can be repeated

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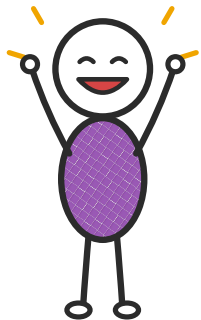
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⚡ THE TRIGGER—GenAI agents arrive. Unpredictable, emergent, co-created artifacts enter the pipeline—and the story begins.

Chapter 1—The substrate changes

3.A—GenAI as Substrate for New Game Artifacts

GenAI brings a truly new kind of game artifact—NPCs with stable personalities, worlds that adapt, stories that emerge.



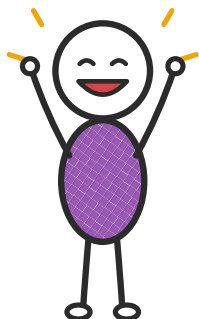
Anchors: Gallotta et al. 2024; Sweetser 2024; Kanervisto et al. 2025

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- Prompts and constraints become real engineering artifacts: versioned, reviewed, maintained



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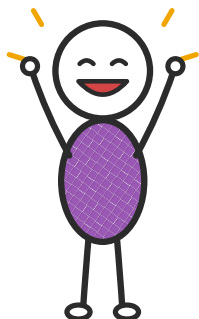
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⚡ Tension—Accountability & opacity

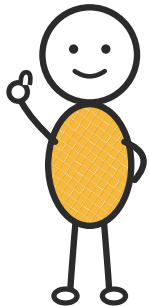
Because content is co-created at runtime, decisions are partly opaque: when an NPC says something offensive or wrong, it is hard to say who is responsible.



Chapter 2—The designer follows

3.B—The Shift in the Designer’s Role

The designer stops writing the story line by line, and instead sets the conditions for the story to emerge during play.



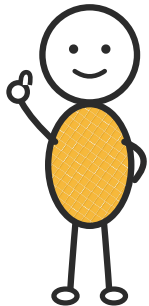
Anchors: Alharthi 2025; Panchanadikar & Freeman 2024; Ling et al. 2024

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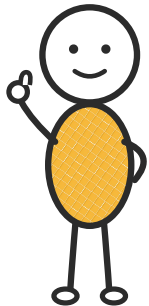
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✓ **Already observed in practice**

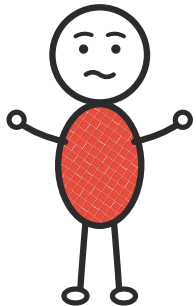
Indie developers describe working like curators: they now iterate on prompts and personality profiles instead of dialogue text.



Chapter 3—The engineer follows

3.C—The Shift in the Engineer's Role

The engineer builds the runtime systems that turn the designer's intent into emergent behavior.



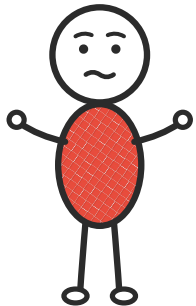
Anchors: Gröpler et al. 2025 (GENIUS); Casamayor et al. 2025; Roca et al. 2024

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The engineer builds the runtime systems that turn the designer's intent into emergent behavior.

- From coding state machines and behavior trees to writing prompts and designing constraints
- Hybrid architectures where classical code and generative models work together
- Addition, not replacement: classical logic stays where it is cheaper, safer, and more predictable



Chapter 3—The engineer follows

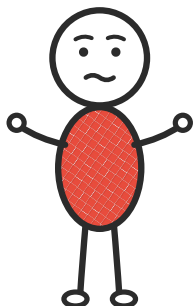
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⚡ Tension—Reproducibility & verifiability

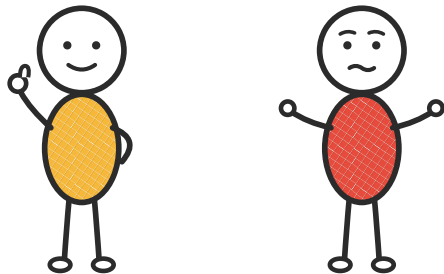
Random output breaks the tests, debugging, and regression checks that classical SE relies on.





3.D—Generative Playtesting: the SE–HCI Convergence

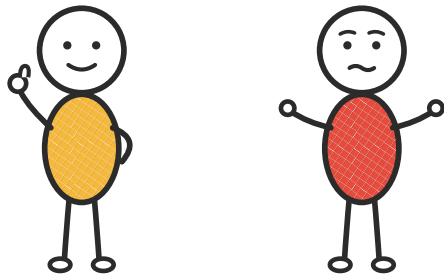
GenAI agents become a shared base for SE-style coverage testing and HCI-style playtesting—inside the same development loop.



3.D—Generative Playtesting: the SE–HCI Convergence

GenAI agents become a shared base for SE-style coverage testing and HCI-style playtesting—inside the same development loop.

- **A new, 8th LLM role in games: the agent as tester, right at the SE–HCI boundary**
- HCI side: LLM agents mimic how differently humans perceive difficulty (Xiao & Yang)
- SE side: LLM agents with different personalities maximize code and interaction coverage (MIMIC)
- The big flip: from tests with one right answer to **distributions of play sessions**



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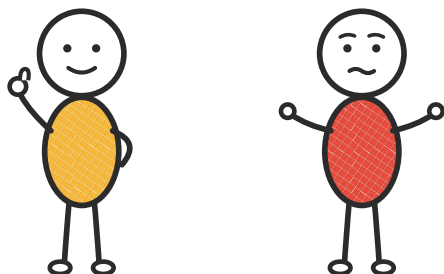
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⚡ Tension—False confidence

Agents are right on average but miss edge cases—outliers and accessibility. The fix: humans and agents testing together.

⚡ Tension—Cost & sustainability

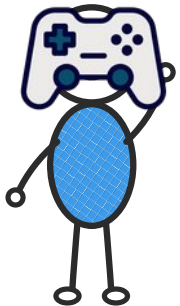
Running models at scale is expensive, and not everyone can afford it; big studios benefit most.



Chapter 5—Labor rearticulates

3.E—Hybrid Roles and the Redistribution of Labor

These shifts don't just rename roles—they reshape the work and give hybrid professionals a home.

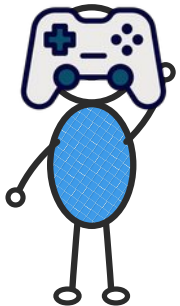


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- designer–engineer hybrids
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Chapter 5—Labor rearticulates

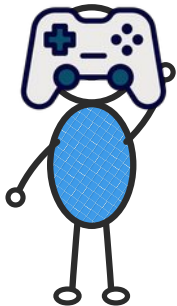
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⚡ **Tension—Deskilling & value redistribution**

Companies can still fire anyone because none likes to deal with humans



Authored

3.A Artifacts

3.B Designer

3.C Engineer

3.D Playtesting

3.E Roles

Convergent

Epilogue

The Convergent Generative Game

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One loop

SE-style coverage testing and HCI-style playtesting run on the same GenAI agents, together inside each development cycle.

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New roles

Hybrid professionals—prompts, personas, constraints, evaluation—get a recognized home; the work becomes about judgment, not mechanics.

Epilogue

The Convergent Generative Game

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SE-style coverage testing and HCI-style playtesting run on the same GenAI agents, together inside each development cycle.

New roles

Hybrid professionals—prompts, personas, constraints, evaluation—get a recognized home; the work becomes about judgment, not mechanics.

One community

Citing across fields, validating distributions, sharing a vocabulary: SE returns to games research as a core contributor.

Credits

Image credits

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6 Game-controller icon—Flaticon; heart-with-pulse icon—free PNG clipart (“Health PNG Image”); bust of Parmenides of Elea—PNGkey; atom clipart—Pngtree

6–11, 30–45, 71–89 Stick figures—author’s own set

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10 First pages of Aarseth 2001, Eskelinen 2001 and Juul 2001 (Game Studies 1(1))

11 First pages of Blow 2004 (ACM Queue), Kanode & Haddad 2009 (ITNG) and Petrillo et al. 2009 (ACM Computers in Entertainment)

12 “King Ghidorah in a Nutshell”—Michael J. Larson (DeviantArt, 2019), via r/GODZILLA

14 Dark figure in the fog—Freepik

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